

EmpathyGym — Product Documentation

What Is EmpathyGym?

EmpathyGym is a **voice-first customer support training simulator** that runs entirely in the browser. A trainee speaks to an AI-powered customer persona in real time, receives live feedback on the customer’s frustration level, and gets a scored performance report at the end of every session — all without any backend server or subscription cost.

The Problem It Solves

Traditional customer support training relies on:

- Role-play with a human trainer (expensive, hard to scale)
- Reading manuals and watching videos (passive, no practice)
- Live calls with real customers (high risk for new agents)

EmpathyGym replaces all three with an **always-available, judgment-free practice environment** where agents can fail safely, learn from mistakes, and build empathy muscle memory before ever touching a real customer.

Use Cases

1. New Agent Onboarding

A call centre hires 20 new agents. Instead of pairing each with a senior trainer for role-play, every agent independently completes 10 EmpathyGym sessions covering different personas before their first live call. Training time drops from days to hours.

2. Empathy Skill Remediation

A QA team flags an agent whose CSAT scores are consistently low. The manager assigns specific EmpathyGym personas that mirror the agent’s weak spots (e.g., “Repeat Caller — Billing Error”). The agent practices until their Empathy Score consistently exceeds 75%.

3. Product Launch Preparation

A company is launching a new product and expects a surge of confused or frustrated callers. Support leads create custom personas matching anticipated complaint types and run the whole team through targeted sessions the week before launch.

4. Self-Directed Continuous Learning

Experienced agents use EmpathyGym on their own to sharpen skills, try harder difficulty personas, and compete on the leaderboard — turning skill development into a habit rather than a one-time event.

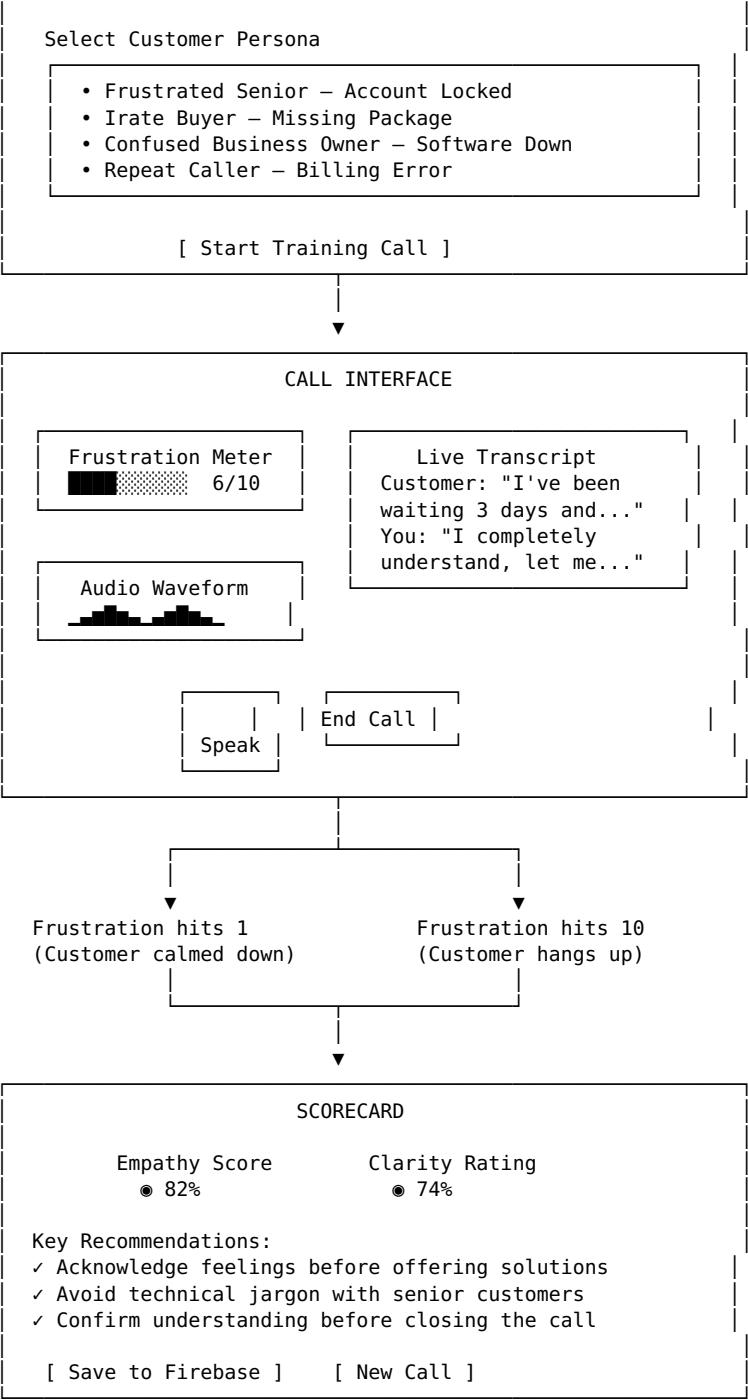
5. Remote / Distributed Teams

Teams spread across time zones can train asynchronously. No trainer needs to be online. Sessions are logged to Firestore so managers can review scores and transcripts at any time.

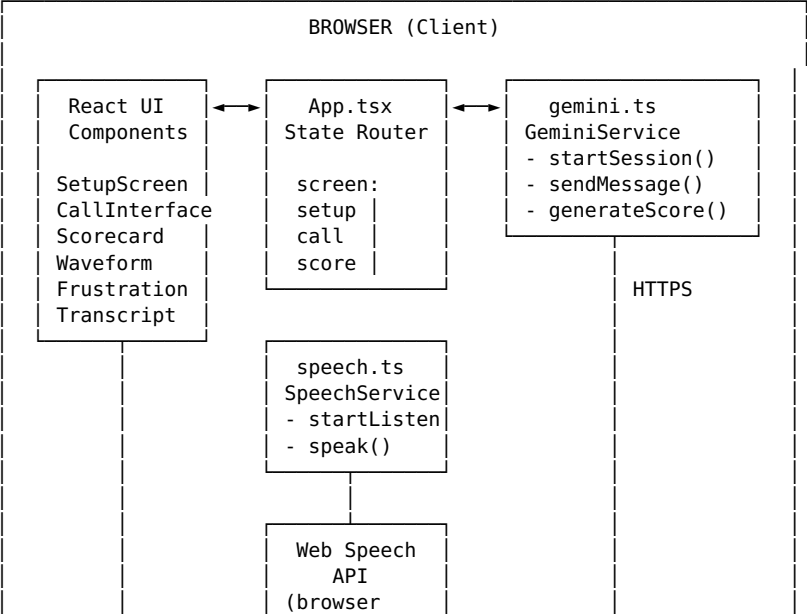
Key Benefits

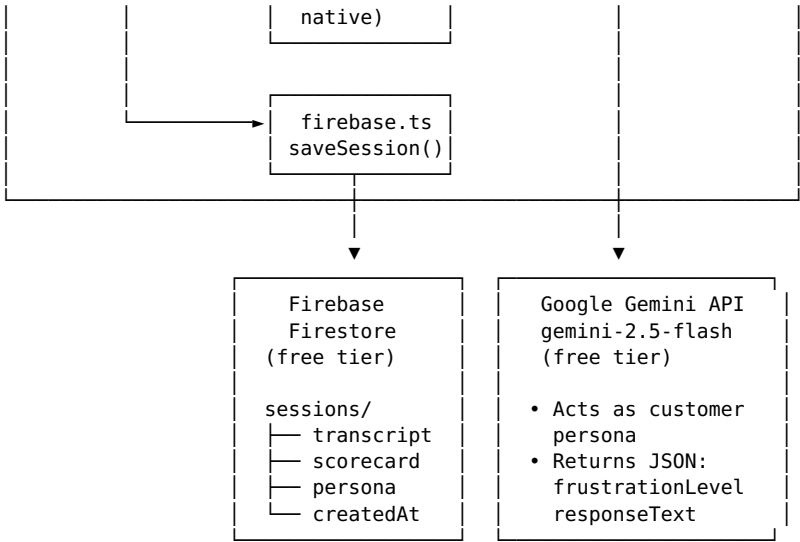
Benefit	Detail
Zero cost to run	Browser-native speech APIs + Gemini free tier + Firebase free tier = \$0/month
Instant feedback	Frustration meter updates after every agent response, not just at the end
Safe to fail	Agents can hang up, restart, and try again with no real-world consequences
Scalable	1 agent or 1,000 agents — no infrastructure to provision
Objective scoring	AI-generated scores remove trainer bias
Async & self-paced	No scheduling, no waiting for a trainer

User Flow

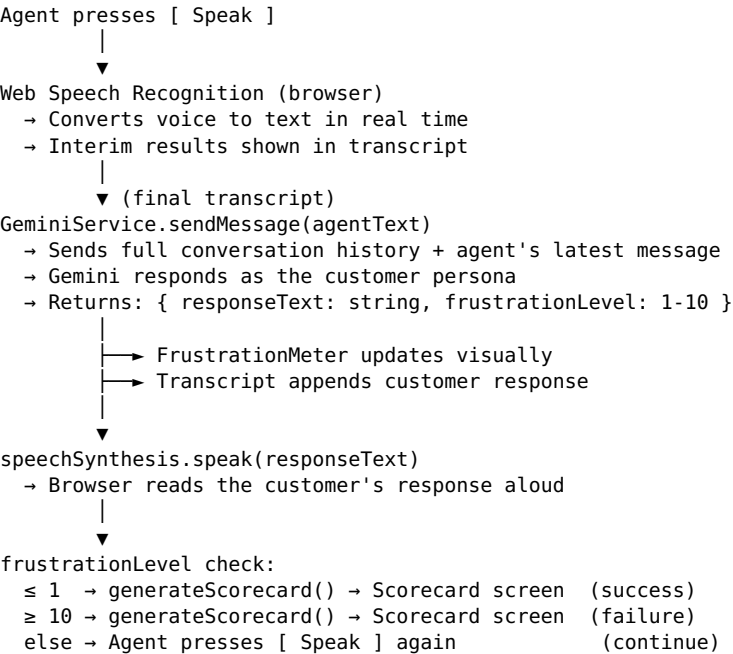


Architecture





Data Flow — Single Turn



Scoring Rubric (AI-Generated)

At session end, the full transcript is sent to Gemini with this evaluation prompt:

Metric	What It Measures
Empathy Score (0-100%)	Did the agent acknowledge feelings, apologise sincerely, and validate the customer’s frustration?
Clarity Rating (0-100%)	Were responses clear, jargon-free, and solution-focused?
Recommendations	3 specific, actionable coaching tips based on the actual transcript

Tech Stack Summary

Layer	Technology	Cost
UI Framework	React 18 + TypeScript + Vite	Free
Styling	Custom CSS (dark theme)	Free
Icons	Lucide React	Free
Speech-to-Text	Web Speech API (webkitSpeechRecognition)	Free (browser native)
Text-to-Speech	Web Speech API (speechSynthesis)	Free (browser native)
AI Reasoning	Google Gemini 2.5 Flash	Free tier (1,500 req/day)
Database	Firebase Firestore	Free tier (50k reads/day)
Hosting	Vercel / Netlify / Firebase Hosting	Free

Total monthly cost: \$0

Deployment

Vercel / Netlify

1. Push repo to GitHub
2. Connect repo in Vercel/Netlify dashboard
3. Set environment variable: VITE_GEMINI_API_KEY=your_key
4. Build command: npm run build · Output: dist

Firebase Hosting

```
npm install -g firebase-tools
firebase login
firebase init hosting    # public dir: dist, SPA: yes
npm run build
firebase deploy
```

Future Enhancements

- **Custom personas** — managers define their own customer scenarios
- **Leaderboard** — team rankings by empathy score pulled from Firestore
- **Session replay** — play back the audio + transcript of any saved session
- **Multilingual support** — Web Speech API supports 50+ languages
- **Manager dashboard** — aggregate scores and trends across the team